Hints: Linear algebra – 4. Functions of matrices and matrix equations

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. A is a perfect square of a simpler integer matrix: $A = B^2$ with $B = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}$. Then $A^n - E = (B^n)^2 - E$.

Hint 2. By induction, $B^n = \begin{pmatrix} p & q \\ 2q & p \end{pmatrix}$ with $p^2 - 2q^2 = \det B^n = (-1)^n$. Compute $B^{2n} - E$: its entries are $p^2 + 2q^2 - 1$, $2pq$, $4pq$.

Hint 3. If n is even, $p^2 + 2q^2 - 1 = 4q^2$, so all entries are divisible by $2q$. If n is odd, $p^2 + 2q^2 - 1 = 2p^2$, so they are divisible by $2p$. Since $p, q \to \infty$, so does d_n.

Problem 2

Hint 1.

(a) What is $f(O, B)$? (b) Which invariant do AB and BA always share? For instance, is (E, O) in the image?

Hint 2.

(a) No, since $f(O, B) = (O, O)$ for every B. (b) No, since $\operatorname{tr} AB = \operatorname{tr} BA$. (c) A fixed point means $AB = A$ and $BA = B$. Transpose the first equation using $A^T = -A$, $B^T = -B$.

Hint 3. You get $BA = -A$, hence $B = -A$, hence $A^2 = -A$. But $A^2 = -A^T A$ is symmetric while $-A$ is skew, so $A^2 = O$, i.e. $A^T A = O$, so $A = O$. The only fixed point is (O, O).

Problem 3

Hint 1. $A^2 = O$ means $\operatorname{im} A \subseteq \ker A$. Combine with $\dim \ker A + \operatorname{rank} A = n$.

Problem 4

Hint 1. Describe S: a symmetric rank-one matrix with entries ± 1 is $\varepsilon u u^T$ with $u \in \{\pm 1\}^{2025}$ and $\varepsilon = \pm 1$. Count $|S|$ carefully (u and $-u$ give the same matrix; the diagonal shows ε).

Hint 2. For $A = \varepsilon u u^T$, $B = \delta w w^T$: $AB = \varepsilon\delta(u \cdot w) u w^T$ and $BA = \varepsilon\delta(u \cdot w) w u^T$. Since 2025 is odd, $u \cdot w$ is an odd integer, never 0.

Hint 3. So A and B commute if and only if $u w^T = w u^T$, i.e. $w = \pm u$, i.e. $B = \pm u u^T$: 2 matrices out of $|S| = 2^{2025}$. The probability is 2^{-2024}.

Problem 5

Hint 1. Let $P = \frac{1}{r} \sum M_i$. Since G is a group, $M_j P = P$ for every j. What does this give for P^2?

Hint 2. $P^2 = P$, and for an idempotent $\operatorname{rank} P = \operatorname{tr} P = 0$.

Problem 6

Hint 1. All the prime entries are non-zero, so the rank is at least 1. Suppose $\operatorname{rank} A = 1$, i.e. $a_{ij} = u_i v_j$ with rational u_i, v_j. Pick one entry for each of $m + n$ distinct primes.

Hint 2. Form a bipartite graph whose vertices are the m rows and n columns, with an edge for each chosen entry. It has $m + n$ vertices and $m + n$ edges, so it contains a cycle $i_1 j_1 i_2 j_2 \dots i_k j_k i_1$.

Hint 3. For a rank-one matrix, $\prod_s a_{i_s j_s} = \prod_s u_{i_s} v_{j_s} = \prod_s a_{i_{s+1} j_s}$. The two products run over complementary halves of the cycle's edges, so in absolute value they are products of disjoint sets of primes. This contradicts unique factorisation.

Problem 7

Hint 1. For "only if", take $P(x) = x^2$ and expand.

Hint 2. For "if", by linearity it suffices to treat $P = x^k$: prove $(A + B)^k = A^k + kA^{k-1}B$ by induction. Multiply by $A + B$ on the right and use $BA + B^2 = AB$.

Problem 8

Hint 1. Show that both sides are positive definite solutions X of the equation $XA^{-1}X = B$, and that this equation has only one positive definite solution (the "geometric mean" of A and B).

Hint 2. For the left side $X = \sqrt{A}\, M \sqrt{A}$ with $M^2 = \sqrt{A}^{-1} B \sqrt{A}^{-1}$, a direct computation gives $XA^{-1}X = B$. The right side Y satisfies $YB^{-1}Y = A$ by the same computation, and inverting gives $YA^{-1}Y = B$.

Hint 3. Uniqueness: if $XA^{-1}X = B$ with X positive definite, then $Z = \sqrt{A}^{-1} X \sqrt{A}^{-1}$ is positive definite with $Z^2 = \sqrt{A}^{-1} B \sqrt{A}^{-1}$. So Z is the unique positive definite square root.

Problem 9

Hint 1. Rewrite the recursion as $c\, x_j = j\, x_{j+1} + (n - j + 1)\, x_{j-1}$ for $j = 1, \dots, n$, with $x_0 = x_{n+1} = 0$. So $(x_1, \dots, x_n)$ is an eigenvector of an $n \times n$ tridiagonal matrix with eigenvalue c, and you want its largest eigenvalue.

Hint 2. Encode the vector as a polynomial $g(t) = \sum_j x_j t^{j-1}$ of degree $\leqslant n - 1$. The eigen-equation becomes the differential equation $(1 - t^2)g' + (n - 1)t\, g = c\, g$.

Hint 3. Try $g = (1 + t)^a (1 - t)^b$ with $a + b = n - 1$: this gives $c = a - b$. The largest value is $c = n - 1$, with $g = (1 + t)^{n-1}$. Answer: $x_k = \binom{n-1}{k-1}$.

Problem 10

Hint 1. Lie–Trotter: let $X_n = e^{A/n} e^{B/n}$ and $Y_n = e^{(A+B)/n}$. Then $\|X_n - Y_n\| = O(1/n^2)$, and both norms are at most $e^{(\|A\| + \|B\|)/n}$. Telescope: $X^n - Y^n = \sum_k X^k (X - Y) Y^{n-1-k}$.

Hint 2. BCH: expand $e^{tA} e^{tB} = E + t(A + B) + t^2 (\frac{A^2}{2} + AB + \frac{B^2}{2}) + O(t^3)$ and apply $\log(E + Z) = Z - \frac{Z^2}{2} + \dots$. The t^2 term is $\frac{1}{2}[A, B]$.

Hint 3. $\det e^A = (\det e^{A/n})^n$, and $\det(E + \frac{A}{n} + O(n^{-2})) = 1 + \frac{\operatorname{tr} A}{n} + O(n^{-2})$; let $n \to \infty$. (Or triangularise: the diagonal of e^A is e^{λ_i}.) Invertibility: $e^A e^{-A} = E$, since A and $-A$ commute.

Problem 11

Hint 1. Look for a symmetry of M. Compare M^T with M after swapping the first two indices.

Hint 2. With P the permutation matrix of the transposition $(1\,2)$, one has $M = PM^TP$. Hence $M^n = P(M^n)^TP$, and comparing diagonal entries gives $(M^n)_{11} = (M^n)_{22}$.

Problem 12

Hint 1.

(a) The equation is $(A + \varepsilon E)X = B$. It is uniquely solvable when $-\varepsilon$ is not an eigenvalue of A; take a smaller than the modulus of every non-zero eigenvalue.

Hint 2.

(b) $\varepsilon \operatorname{tr}(BX) = \varepsilon \operatorname{tr}\big(B(A+\varepsilon E)^{-1}B\big) = \varepsilon \operatorname{tr}(A+\varepsilon E)^{-1}$, using the cyclicity of the trace and $B^2 = E$. This equals $\sum_i \frac{\varepsilon}{\lambda_i + \varepsilon}$.

Hint 3. The limit counts the zero eigenvalues of A with algebraic multiplicity. For diagonalisable A this equals $\dim\ker A = n - \operatorname{rank} A$.

Problem 13

Hint 1. If $p(B) = a_0E + \cdots + a_mB^m$ is invertible, then $A = Bp(B)^{-1}$ is a function of B and commutes with it. So show that $p(B)$ is invertible.

Hint 2. Otherwise $p(\mu) = 0$ for some eigenvalue μ of B, and $\mu \neq 0$ since $p(0) = a_0 \neq 0$. Apply the relation to an eigenvector $Bv = \mu v$: the left side gives $p(\mu)Av = 0$, the right side gives $\mu v \neq 0$.

Problem 14

Hint 1. (Real matrices.) Guess the solutions from the scalar case $y = 1/(c - x)$: check that $Y(x) = (C - xE)^{-1}$ solves $Y' = Y^2$.

Hint 2. If $Y(x_0)$ is invertible, put $C = Y(x_0)^{-1} + x_0E$. By uniqueness for the ODE, $Y = (C - xE)^{-1}$ wherever the solution exists. It is defined on all of $\mathbb{R}$ if and only if C has no real eigenvalue.

Hint 3. A real matrix of odd size always has a real eigenvalue; one of even size need not (block rotations). Answer: (i) yes, (ii) no.